



Figure 8: For Flower dataset, average top- k intersection, 1-LENS-prec@ k , 1-LENS-recall@ k measured between IG(original image) and IG(perturbed image) for models that are naturally trained, PGD-trained and IG-SUM-NORM trained. The perturbation used is the top- t attack of (?). Note top- k is equivalent to 0-LENS-prec@ k , 0-LENS-recall@ k .

1-LENS-Spearman and 1-LENS-Kendall measures to quantify the attributional robustness to the top- k attack of ?, and it holds across the datasets used by ? in their study; see Appendix ??.

? show theoretically that ℓ_∞ -adversarial training (PGD-training) leads to stable Integrated Gradients (IG) under ℓ_1 norm. They also show empirically that PGD-training leads to sparse attributions (IG & DeepSHAP) when sparseness in measured indirectly as the change in Gini index. Our empirical results extend their theoretical observation about stability of IG for PGD-trained models, as we measure local stability in terms of both the top attribution values and their positions in the image.

Table 2 obtains the top- k intersection, 3-LENS-recall@ k , and 3-LENS-recall@ k -div of different attribution methods on ImageNet for naturally trained and PGD-trained ResNet50 models. We observe that for random sign attack the improvement obtained on top- k intersection is reduced for a large dataset like ImageNet. Still our conclusions about locality and diversity in attribution robustness in comparison with the top- k intersection baseline holds as we observe improvements in using diversity and locality. More results about incorporating diversity in the attribution and the resulting robustness metrics are available in Appendix ??.

Training	Attribution method	top- k	3-LENS-recall@ k	3-LENS-recall@ k -div
Natural	Simple Gradient	0.3825	0.7875	0.8290
Natural	Image $\times$ Gradient	0.3316	0.7765	0.8655
Natural	LRP [Bach 2015]	0.1027	0.2487	0.7518
Natural	DeepLIFT [Shrikumar 2017]	0.2907	0.7641	0.8504
Natural	GradSHAP [Lundberg 2017]	0.2290	0.6513	0.8099
Natural	IG [Sundararajan 2017]	0.2638	0.7148	0.8380
PGD	Simple Gradient	0.1725	0.7245	0.8004
PGD	Image $\times$ Gradient	0.1714	0.7269	0.8552
PGD	LRP [Bach 2015]	0.2374	0.4147	0.8161
PGD	DeepLIFT [Shrikumar 2017]	0.5572	0.9746	0.8977
PGD	GradSHAP [Lundberg 2017]	0.1714	0.7270	0.8552
PGD	IG [Sundararajan 2017]	0.1947	0.7335	0.8584

Table 2: Average top- k intersection, 3-LENS-prec@ k and 3-LENS-prec@ k -div for random sign perturbation attack applied to different attribution methods on ImageNet for naturally and adversarially(PGD)-trained ResNet50 models.

6 Conclusion and Future Work

We show that the fragility of attributions is an effect of using fragile robustness metrics such as top- k intersection that only look at the rank order of attributions and fail to capture the locality of pixel positions with high attributions. We highlight the need for locality-sensitive metrics for attributional robustness and propose natural locality-sensitive extensions of existing metrics. We introduce another method of picking diverse top- k pixels that can be naturally extended with locality to obtain improved measure of attributional robustness. Theoretical understanding of locality-sensitive metrics of attributional robustness, constructing stronger attributional attacks for these metrics, and using them to build attributionally robust models are important future directions.

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